Category: OPTIMIZATION | Speedup: Polynomial | Concept ID #35

Foundational Research & Advances in Tensor Networks & Matrix Product States (MPS)

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1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Tensor Networks & Matrix Product States (MPS) in modern quantum information architectures.

2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Instead of storing a billion puzzle pieces that connect everything to everything, you line them up like train cars connected only to their immediate neighbors, saving massive memory."

3. Mathematical Formulation & Unitary Rule

$$|\psi\rangle = \sum_{i_1, \dots, i_n} \text{Tr}(A^{[1], i_1} A^{[2], i_2} \dots A^{[n], i_n})$$

Matrix Product State tensor decomposition compresses n-body quantum states to polynomial O(n * d * chi^2) memory.

4. Step-by-Step Circuit Sequence

Sequential contraction of rank-3 tensors along 1D chain; singular value decomposition (SVD) truncation per two-qubit gate.

5. Executable IBM Qiskit (Python) Code

```python
from qiskit import QuantumCircuit
from qiskit_aer import AerSimulator

# 100-qubit Greenberger-Horne-Zeilinger (GHZ) state with MPS
qc = QuantumCircuit(100)
qc.h(0)
for i in range(99):
    qc.cx(i, i+1)
```

6. Computational Complexity & Speedup

Classical Time:	Quantum Time:	Speedup Class:
O(n * chi^3) for MPS state contraction vs O(2^n) full statevector	Natural physical representation on 1D quantum spin chains	Polynomial

7. Key Applications & Future Research Directions

- Simulating 100+ qubit shallow circuits classically (DMRG, cuTensorNet)
- Quantum-inspired machine learning via Tensor Network architectures
- Benchmark verification for NISQ hardware quantum advantage experiments

Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Optimization pipelines.